UNIVERSITY OF LOUISVILLE
J.B. SPEED SCHOOL OF ENGINEERING

Department of Industrial and Systems Engineering

Fully-Funded Ph.D. Positions (up to 2)
MINDxAI Lab, University of Louisville

Position: Ph.D. Student
Start Date: Fall 2026 or Spring 2027

The **Modeling Interaction, cogNition, and Decision-making by AI (MINDxAI)** Lab at the University of Louisville is expanding and will offer **up to two fully funded Ph.D. positions** in the Department of Industrial and Systems Engineering. The selected students will work under the supervision of **Dr. Yunmei Liu**. These positions focus on adaptive systems design, with an emphasis on human-in-the-loop intelligent systems that integrate artificial intelligence/machine learning (AI/ML) to improve system efficiency, safety, and robustness. This opportunity is well suited for highly motivated students interested in applying AI/ML techniques to real-world human-centric AI systems.

Current lab projects span aviation, transportation, and healthcare, with cross-domain and cross-institutional collaborations. Ongoing research projects include:

(1) Cognitive-Emotional State Modeling for Real-Time Adaptive Automation (NSF-funded). This research develops AI models for real-time inference of human cognitive-emotional states from multimodal physiological and behavioral data. These AI models serve as the decision foundation for adaptive automation systems, enabling data-driven determination of when and how automation should intervene in safety-critical driving contexts.
(2) Aggressive Driving in Mixed Human-Autonomous Vehicle Traffic. This research leverages ML and data-driven modeling to characterize aggressive driving behaviors and associated human states in mixed human–automated traffic. AI is used to link individual-level behavioral patterns to system-level outcomes, supporting the design of human-aware and adaptive automated driving behaviors.
(3) Reliable Seizure Prediction with False-Alarm Control. This project focuses on developing robust AI models for EEG-based seizure detection and forecasting under realistic conditions. The work emphasizes AI reliability, generalization across domains, and false-alarm control, with the goal of enabling deployable, safety-critical seizure warning systems.

Incoming students should be proficient in AI/ML applications and eager to address real-world, safety-critical problems through interdisciplinary research.

Qualifications:
1. A Bachelor's degree or, preferably, a Master's degree in Industrial and Systems Engineering, Biomedical Engineering, Electrical and Computer Engineering, Computer Science, or a closely related field.
2. Experience or strong interest in AI/ML and human-subjects experimentation.
3. Strong English writing and communication skills.

Interested candidates are encouraged to contact **Dr. Yunmei Liu** at yunmei.liu@louisville.edu. Please use the subject line "PhD position: [Your Name]" and include your CV and transcripts in the email.

About Dr. Yunmei Liu: Dr. Liu is an Assistant Professor in the Department of Industrial and Systems Engineering at the University of Louisville. She directs the **MINDxAI Lab** and leads the **Human-Centric AI** theme at the Center for Human Systems Engineering. She has received research funding from the National Science Foundation (NSF) as Leading Principal Investigator to support her lab.

Her research focuses on cognitive workload modeling, human-centric AI, and human–system co-adaptive design, with an emphasis on developing adaptive systems that respond to dynamic human states. Her work has been published in leading venues across **both industrial and systems engineering** (e.g., *IEEE Transactions on Human-Machine Systems*, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *International Journal of Human-Computer Studies*, *Ergonomics*, *Human Factors*, and *Applied Ergonomics*) **and AI** (e.g., *International Conference on Learning Representations*). She currently serves as an **Associate Editor** for *IEEE Transactions on Human-Machine Systems*.

About the Department: The Department of Industrial and Systems Engineering is within the Speed School of Engineering located on the Belknap campus. The department has been a degree granting department since the late 1970s. The department has 13 faculty with the following research thrusts: operations research and decision analytics, human factors and ergonomics, advanced manufacturing, and engineering education. Faculty research has been supported by a variety of federal grants and industry contracts (e.g., NSF, DoD, ONR, NIST, NASA, FHWA, Toyota, GE Appliances/Haier).

About the MINDxAI Lab:

The **MINDxAI Lab**, founded and directed by Dr. Liu, is located in Henry Vogt Building, Room 312. The lab is equipped with state-of-the-art facilities for human-centric AI and adaptive systems research, including a driving simulator, EmotiBit sensors, eye-tracking systems, EEG headsets, and Delsys electromyography sensors. To support the computational demands of large-scale physiological data processing, the lab maintains substantial high-performance computing resources, including **four B200 GPUs**, **20 active CPUs (with an additional 180 in queue)**, and **4 TB of dedicated storage**.